Operating Systems Project Explanation (Input Format & Dataset Variants)

1. Input File Format

The project requires reading CSV files from an input directory. Each file contains multiple records, where each line represents one record and values are comma-separated.

Structure:
- First column = Key (depends on dataset variant)
- Remaining columns = Numeric values to process

Important:
- Do not hardcode columns or values
- Each file may have different number of columns but is internally consistent

Example:

Electronics,100,200,150
Clothing,50,60,70

2. Role of Input in Pipeline

The CSV data flows through different components:

Ingester:
- Reads CSV files
- Splits into chunks
- Sends chunks via FIFO

Processor:
- Uses threads to process rows
- Extracts key and values
- Performs aggregation

Reporter:
- Reads final data from shared memory
- Writes report.txt and report.csv

3. Dataset Variants

Each student group is assigned one dataset variant. You only implement logic for that variant.

A. Retail Transactions

Goal:
- Calculate total revenue per category
- Find top-N products

Example:
Electronics,100,200,150

Logic:
Sum all values per category

B. IoT Sensor Logs

Goal:
- Calculate moving averages
- Detect anomalies

Example:
Sensor1,20,22,25,100

Logic:
Process readings and identify unusual values

C. Web Clickstream

Goal:
- Calculate session length
- Calculate bounce rate

Example:
User1,5,10,3
User2,1

Logic:
Single value = bounce

D. Financial Tick Data

Goal:
- Calculate VWAP
- Find high and low values
- Calculate total volume

Example:
AAPL,150,100
AAPL,155,200

4. Full Pipeline Flow

CSV Files → Ingester → FIFO → Processor (Threads) → Shared Memory → Reporter

5. Key Learning Objective

This project focuses on Operating Systems concepts:

- Process creation (fork, exec)
- Threads (pthread)
- Inter-process communication (FIFO, shared memory)
- Synchronization (mutex, semaphores)
- Signals handling

We read CSV files where the first column is the key and remaining columns are numeric values. Based on the assigned dataset variant, aggregation is performed using a multithreaded processor, and results are passed via shared memory to the reporter for final output.